

Name: Villondo, Angel Rain L.

Course: Numerical Methods in Civil Engineering

Date: August 25, 2026

Write a complete, standalone Python script (lab02_villondo.py) that analyzes the reservoir stage dataset located at C:\Users\AngelV\Downloads\Data 01.1.xlsx and outputs an interactive HTML dashboard named lab02_villondo.html.

Data Processing Requirements:

- Read the Excel file starting at row index 3, clean the timestamps, and derive elapsed hours ($t = 0.00, 0.25, \dots$).
- Compute 1st ($\frac{dh}{dt}$) and 2nd ($\frac{d^2h}{dt^2}$) finite-difference derivatives, identifying peak inflow rate and its corresponding timestamp.
- Fit a 4-parameter logistic curve $h(t) = c + \frac{a}{1 + e^{-k(t-t_0)}}$ using `scipy.optimize.curve_fit` (Levenberg-Marquardt). Ensure array NaN values cleanly convert to JSON null.
- Compute statistical metrics (SSE, SST, R^2, s), parameter standard errors, t-statistics, and two-tailed p-values.
- Compute stage volume integration using `scipy.integrate.quad` on the fitted curve and validate against `np.trapezoid`.

Dashboard UI & Design Requirements:

- Theme: Light Executive Technical Report style featuring a pure white background (#ffffff), soft slate card containers (#f8f9fc), dark navy typography (#0f172a), and clean slate borders (#e2e8f0).
- Layout: Top header block with project metadata and a 4-column KPI highlight grid.
- Navigation: 3-tab layout:
 1. *Numerical Derivatives & Kinematics*
 2. *Logistic Model & Curve Fitting*
 3. *Statistical Diagnostics & Volumetric Integration*

- Visualizations & Data: High-contrast Chart.js interactive charts using blue, teal, and amber tones (#0284c7, #0d9488, #d97706). Include styled parameter summary tables with hover effects and badge tags.